



LAB3

CONDITIONALS

Recommended exercises

- 3.1.2
- 3.2.2
- 3.2.4
- 3.2.9

Exercise 03.1. 2

- **As always, read the request on the PDF first...**

Exercise 03.1. 2

■ Examples :

Input : “Horse”

Output:

Contains only letters

Starts with a capital letter

Input : “C1P8.”

Output:

Starts with a capital letter

Ends with a period

Exercise 03.1. 2

- **Tips:** Python provides several functions to check whether the characters in a string are:

- Alphanumeric : *"hello..."*. `isalnum()`
- Capital : *"world"*. `isupper()`
-

A complete list can be found here

https://www.w3schools.com/python/python_ref_string.asp

- NB: The program must work for strings of **any** length

Exercise 03.2. 2

- **As always, read the request on the PDF first...**

Exercise 03.2. 2

- We have to write a program that asks for a grade composed of a letter and a “+” or “-” sign. Once entered, the grade must be converted into a numeric value **depending on the letter entered** . Then you must subtract or reduce 0.3 from the number **depending on the sign** . Finally, print the numeric grade obtained.
- For example, if the read grade is “ **D+** ”, then the output value will be **1 + 0.3** , so the value **1.3 should be printed**.

Exercise 03.2.2

- Pay attention to special cases:
 - **F** never has a sign following it
 - The **+** in **A+** has no effect
- We can access single string characters by their index (*string[Index]*):
 - 0 for the letter
 - 1 for The Sign
- Depending on the length of the string we will have to do more or less operations...
- **SUGGESTION:**
 - The letter is definitely there: we process that first and then "refine" it based on the optional modifier (+ or -)

Exercise 03.2. 4

- **As always, read the request on the PDF first...**

Exercise 03.2.4

- Given a year as input, determine whether it is a leap year using the following algorithm.
 - If it is divisible by 4 **and** less than 1582, it is a leap year.
 - If it is divisible by 4 **and** greater than 1582 **and** is not divisible by 100, it is a leap year.
 -

Instead of using multiple **if - elif -else statements** combine all the previous points into a single statement, thanks to the logical operators **and, or, not** .

Exercise 03.2.4

- The **% operator** calculates the remainder of a division.
- To make sure that the program works in all cases, try with different inputs.
 - 1580
 - 1996
 - 2000
 - 1900

Exercise 03.2. 9

- **As always, read the request on the PDF first...**

Exercise 03.2.9(cont'd)

- Refer to the Richter scale exercise done in class
- It involves writing a series of `if....elif...else` clauses.